

(a) Fig. 7.1 shows two chromosomes in one stage of meiosis.

The letters G to M represent the dominant alleles of seven genes and the letters g to m represent the recessive alleles of the same seven genes.

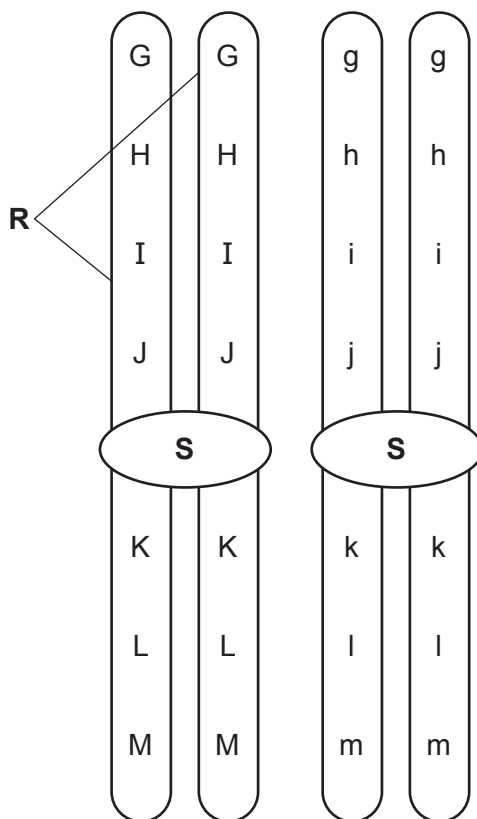


Fig. 7.1

- (i) Name the structures labelled **R** and **S** on Fig. 7.1.

R

S

[2]

- (ii) State **three** features visible on Fig. 7.1 that identify the chromosomes as a homologous pair.

..... [3]

[3]

- (iii) Fig. 7.2 shows the same two chromosomes a little later in the same stage of meiosis. Crossing over is beginning to occur at point T.

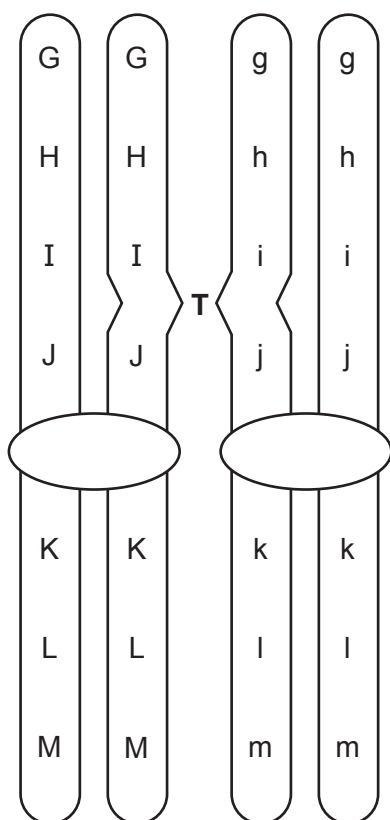


Fig. 7.2

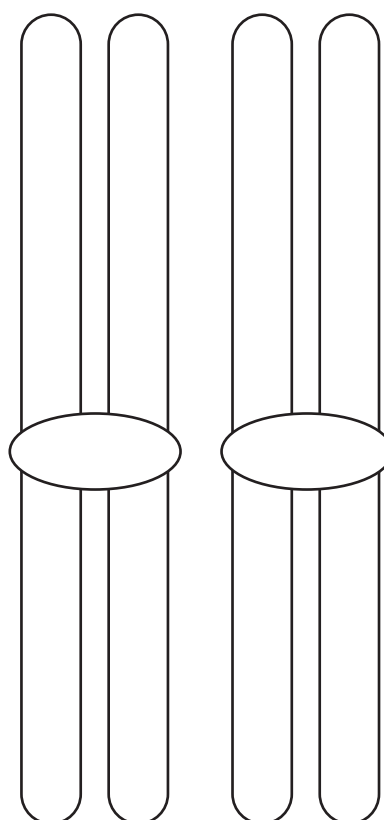


Fig. 7.3

Fig. 7.3 shows an outline of the same two chromosomes after crossing over has occurred.

Complete Fig. 7.3 by writing in the letters of the alleles along both chromosomes. Take care to clearly show the difference between letters representing dominant alleles and letters representing recessive alleles.

[2]

- (b) State the stage in meiosis in which crossing over occurs.

..... [1]

(c) Crossing over results in genetic variation.

Explain how random assortment of homologous chromosomes also results in genetic variation.

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..... [3]

[Total: 11]